SUMMIT

STEM CATEGORY

SLOT 2

STEM ENIGMA

Mathematics & Integrated Sciences Category

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INTRODUCTION

- STEM Enigma challenges students from Grades 8–11 to apply their mathematical reasoning, scientific knowledge and logical problem-solving skills.
- Participants work in mixed-subject teams during the initial round, combining mathematics and integrated sciences to investigate and solve challenging case files. Qualifying participants then advance to an individual scavenger hunt, where they put their subject-specific knowledge and quick thinking to the test.
- This study guide outlines the key concepts and skills required to succeed in both rounds.

OVERVIEW

- The category consists of two main rounds: Investigating Case Files (Team Round) and Scavenger Hunt Style Quiz (Individual Round).
- Participants first register individually for Mathematics or Integrated Sciences and are then grouped into mixed-subject teams to solve a case file using mathematical and scientific knowledge.
- The top-performing teams advance to the final round, where each qualifying participant competes individually in a subject-specific scavenger hunt by solving questions within a given time limit.
- Mathematics and Integrated Sciences have separate competitions, with a Winner and Runner-Up declared for each category.

CONTENT – MATHEMATICS

STUDENTS REGISTERING AS STEM SYNDICATES NEED TO BE ABLE TO:

1. Number Systems and Arithmetic

- Natural numbers, whole numbers, integers and rational numbers
- Positive and negative numbers
- Fractions, decimals and percentages
- Order of operations (BODMAS)
- Factors, multiples, HCF and LCM
- Prime numbers and prime factorisation
- Squares, cubes, square roots and cube roots
- Standard form (scientific notation)
- Ratios and proportions
- Direct and inverse proportion
- Rounding, estimation and significant figures

2. Algebra

- Simplification of algebraic expressions
- Expansion and factorisation

- Algebraic identities
- Substitution into expressions and formulae
- Solving linear equations
- Simultaneous linear equations
- Quadratic expressions and equations
- Sequences and patterns
- Finding the nth term of a sequence
- Inequalities and their representation on a number line

3. Geometry

- Types of angles and their properties
- Angles on a straight line and around a point
- Angle properties (corresponding, vertically opposite angles etc)

4. Mensuration

- Perimeter and area of basic 2D shapes
- Area of triangles, circles, parallelograms and trapeziums
- Circumference of circles
- Surface area and volume of 3D shapes
- Volume of cubes, cuboids, cylinders and prisms

5. Trigonometry

- Pythagoras Theorem
- Using sin, cos and tan to find angles and sides of triangles

6. Graphs and Functions

- Cartesian coordinates
- Plotting and interpreting graphs
- Linear graphs and their gradients
- Finding the equation of a straight line
- Quadratic graphs

7. Statistics and Probability

- Mean, median, mode and range
- Bar charts, pie charts and histograms

8. Mathematical Reasoning and Problem-Solving

- Number and shape patterns
- Logical reasoning
- Identifying missing numbers
- Mathematical riddles
- Applying mathematical concepts to unfamiliar situations
- Mental mathematics and quick calculations

CONTENT – INTEGRATED SCIENCES

STUDENTS REGISTERING AS STEM UNDERWORLD NEED TO BE ABLE TO:

A: BIOLOGY

Not compulsory for non-bio students

1. Cells and Organisation

- Structure and functions of animal and plant cells
- Differences between plant and animal cells
- Functions of cell organelles
- Specialised cells
- Levels of organisation: cells, tissues, organs and organ systems

2. Human Body Systems

- Digestive system and digestion
- Respiratory system and gas exchange
- Circulatory system and blood circulation
- Functions of the heart and blood vessels

3. Photosynthesis

- Photosynthesis and its requirements
- Factors affecting the rate of photosynthesis
- Differences between photosynthesis and respiration

4. Respiration and Transport

- Aerobic and anaerobic respiration
- Differences between aerobic and anaerobic respiration
- Gas exchange in plants and animals
- Transport of water and minerals in plants
- Functions of xylem

5. Reproduction

- Sexual and asexual reproduction
- Plant reproduction and pollination

B: CHEMISTRY

1. States of Matter

- Solids, liquids and gases
- Properties of different states of matter
- Changes of state
- Melting, freezing, evaporation, boiling and condensation
- Particle theory and diffusion

2. Atomic Structure and the Periodic Table

- Structure of atoms
- Protons, neutrons and electrons
- Atomic number and mass number
- Elements, compounds and mixtures
- Arrangement of elements in the periodic table
- Groups and periods
- Basic properties of metals and non-metals

3. Chemical Bonding and Formulae

- Ionic and covalent bonding
- Formation of ions

4. Chemical Reactions

- Reactants and products
- Word equations and symbol equations
- Balancing simple chemical equations
- Types of chemical reactions
- Exothermic and endothermic reactions

5. Acids, Bases and Salts

- Properties of acids and bases
- pH scale
- Indicators and colour changes
- Neutralisation reactions
- Formation of salts
- Common laboratory acids and alkalis

6. Metals and Reactivity

- Physical and chemical properties of metals
- Reactivity series
- Reactions of metals with oxygen, water and acids
- Displacement reactions

7. Mixtures and Separation Techniques

- Filtration
- Evaporation
- Crystallisation
- Distillation
- Fractional distillation
- Chromatography
- Identifying suitable separation techniques

C: PHYSICS

1. Motion and Forces

- Speed, velocity and acceleration
- Distance-time graphs
- Speed-time graphs
- Types of forces
- Effects of forces on motion
- Newton’s laws of motion (basic understanding)
- Mass and weight
- Friction and air resistance
- Momentum (basic understanding)

2. Energy

- Different forms of energy
- Energy transfers and transformations
- Law of conservation of energy
- Kinetic and gravitational potential energy
- Renewable and non-renewable energy resources
- Efficiency and energy wastage

3. Work, Power and Pressure

- Work done
- Power
- Pressure in solids and fluids
- Factors affecting pressure
- Applications of pressure in everyday life

4. Waves, Light and Sound

- Properties of waves
- Transverse and longitudinal waves
- Frequency, wavelength and amplitude
- Reflection and refraction of light

5. Heat and Temperature

- Difference between heat and temperature
- Thermal expansion
- Conduction, convection and radiation
- Changes of state
- Specific heat capacity (basic understanding)